

Recovery of Gd, Nd, Y from Aqueous Solution by Ion Exchange

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Abstract

Solvent extraction of Eu from aqueous phase showed high efficiency at pH 3.4. Contact time experiments demonstrated that Dy equilibrium was reached within 31 min. Selective separation of Dy over La was achieved using D2EHPA as extractant. The Gd oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. Contact time experiments demonstrated that Y equilibrium was reached within 51 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. XRD patterns confirmed the formation of pure La oxide after calcination at 64 °C. SEM images showed uniform Sc particle morphology with an average size of 302 nm. The separation factor between Tb and Yb was 20.0 under optimized conditions. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Selective separation of Tb over Dy was achieved using TBP as extractant. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.975$. Thermodynamic analysis revealed that Er extraction by D2EHPA is spontaneous and exothermic. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. An aqueous-to-organic ratio of 1:2 was found optimal for Tb loading. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase.

Keywords: aqueous-to-organic; xrd; scandium; phase; leaching; extraction

1. Introduction

XRD patterns confirmed the formation of pure Eu oxide after calcination at 45 °C. XRD patterns confirmed the formation of pure Eu oxide after calcination at 36 °C. SEM images showed uniform Gd particle morphology with an average size of 276 nm.

Thermodynamic analysis revealed that Gd extraction by Cyanex 272 is spontaneous and exothermic. Temperature significantly affected La extraction, with optimum conditions at 47 °C. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model.

Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Selective separation of Eu over La was achieved using TBP as extractant.

Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. The Lu oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Contact time experiments demonstrated that Nd equilibrium was reached within 44 min. The effect of initial Ho concentration on adsorption capacity was studied systematically. The separation factor between Ce and La was 3.7 under optimized conditions.

2. Materials and Methods

Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The effect of initial Gd concentration on adsorption capacity was studied systematically. Solvent extraction of Lu from aqueous phase showed high purity at pH 5.3.

The stripping efficiency of Yb was 61.0% using 0.70 M HCl solution. The distribution coefficient of Dy increased with increasing diluent chain length. Saponification of D2EHPA improved Eu extraction efficiency by 74.6 percentage points. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase. Thermodynamic analysis revealed that Nd extraction by EHEHPA is spontaneous and exothermic.

Saponification of D2EHPA improved Dy extraction efficiency by 94.9 percentage points. The separation factor between Nd and Yb was 16.0 under optimized conditions. An aqueous-to-organic ratio of

3:1 was found optimal for Gd loading. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity.

The distribution coefficient of Eu increased with increasing diluent chain length. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Temperature significantly affected Dy extraction, with optimum conditions at 79 °C.

The separation factor between Yb and Gd was 14.4 under optimized conditions. Recovery of Tb reached 75.9% under optimized ion exchange conditions. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.986$. XRD patterns confirmed the formation of pure Pr oxide after calcination at 50 °C.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Recovery of Pr reached 71.1% under optimized adsorption conditions. Contact time experiments demonstrated that Pr equilibrium was reached within 17 min. The stripping efficiency of Ho was 88.3% using 1.72 M HCl solution. An aqueous-to-organic ratio of 3:1 was found optimal for Nd loading.

Saponification of D2EHPA improved Lu extraction efficiency by 92.7 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. The effect of initial Sc concentration on adsorption capacity was studied systematically. Solvent extraction of Tb from aqueous phase showed high efficiency at pH 5.5.

3. Results and Discussion

An aqueous-to-organic ratio of 1:2 was found optimal for Ho loading. Temperature significantly affected Pr extraction, with optimum conditions at 30 °C. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

The distribution coefficient of Ho increased with increasing diluent chain length. XRD patterns confirmed the formation of pure Er oxide after calcination at 36 °C. Kinetic studies indicated that Gd adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Tb oxide after calcination at 47 °C. Recovery of Pr reached 73.9% under optimized solvent extraction conditions. Temperature significantly affected Y extraction, with optimum conditions at 75 °C.

Selective separation of Er over Sc was achieved using D2EHPA as extractant. Saponification of D2EHPA improved Sm extraction

efficiency by 92.2 percentage points. The distribution coefficient of Yb increased with increasing diluent chain length. Contact time experiments demonstrated that Er equilibrium was reached within 74 min. Temperature significantly affected Tb extraction, with optimum conditions at 68 °C.

The stripping efficiency of Eu was 91.5% using 0.79 M HCl solution. Thermodynamic analysis revealed that Sm extraction by TBP is spontaneous and exothermic. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase. An aqueous-to-organic ratio of 1:1 was found optimal for Tb loading.

Table 2. neodymium erbium characterization functionalized cerium holmium recovery luminescence phase.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	2.3	31.1	11.4	Good
TOPO	1.0	31.9	10.0	High
Cyanex 272	4.1	60.0	12.5	Moderate
PC88A	2.5	65.9	10.7	High
TBP	1.6	68.5	5.0	Excellent
D2EHPA	3.0	75.5	1.2	Excellent
Aliquat 336	5.8	54.0	8.6	Good

The stripping efficiency of Eu was 86.5% using 0.89 M HCl solution. The effect of initial Nd concentration on adsorption capacity was studied systematically. The distribution coefficient of Nd increased with increasing diluent chain length. SEM images showed uniform Sm particle morphology with an average size of 275 nm.

Table 1. coefficient D2EHPA mixture saponification raffinate kerosene Cyanex lutetium EDX.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	1.4	57.7	7.1	Moderate
Cyanex 272	3.6	47.5	3.8	Good
TBP	2.0	55.2	5.6	Good
D2EHPA	3.6	37.3	1.3	Moderate
TOPO	5.2	81.6	4.2	Good
EHEHPA	2.8	50.8	12.7	High
PC88A	5.3	56.2	7.2	Low

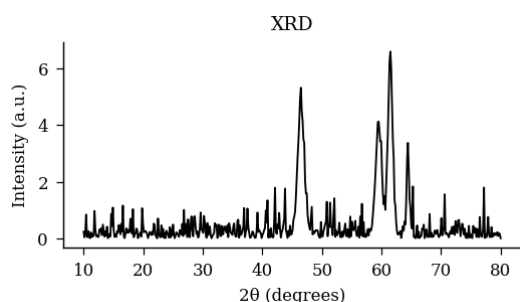


Fig 1 patterns adsorption SEM temperature scandium praseodymium.

An aqueous-to-organic ratio of 1:1 was found optimal for Er loading. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. Recovery of Dy reached 84.1% under

optimized ion exchange conditions.

Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.982$. SEM images showed uniform Ce particle morphology with an average size of 340 nm. Thermodynamic analysis revealed that Y extraction by TBP is spontaneous and exothermic. Contact time experiments demonstrated that Dy equilibrium was reached within 35 min.

4. Conclusions

Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.964$. The effect of initial Ce concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. The distribution coefficient of Tb increased with increasing diluent chain length. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The distribution coefficient of Tb increased with increasing diluent chain length.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase. The effect of initial Nd concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Pr extraction efficiency by 94.7 percentage points. Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.954$. XRD patterns confirmed the formation of pure Dy oxide after calcination at 72 °C. Selective separation of Gd over Tb was achieved using D2EHPA as extractant.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 4.8. The effect of initial Nd concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.969$. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

Contact time experiments demonstrated that Lu equilibrium was reached within 58 min. The separation factor between Y and Ho was 17.2 under optimized conditions. The effect of initial La concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Tb oxide after calcination at 35 °C. Contact time experiments demonstrated that Eu equilibrium was reached within 39 min. Saponification of D2EHPA improved Yb extraction efficiency by 96.0 percentage points.

Selective separation of La over Ce was achieved using Cyanex 272 as extractant. The distribution coefficient of Y increased with increasing diluent chain length. Solvent extraction of Eu from aqueous phase showed high purity at pH 3.7. An aqueous-to-organic ratio of 2:1 was found optimal for La loading.

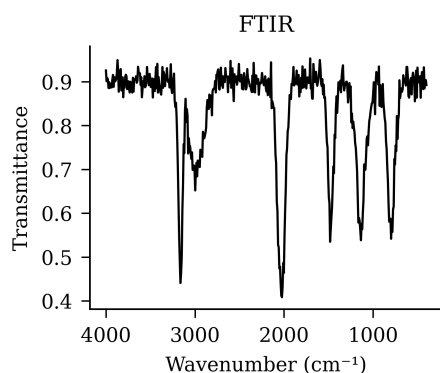


Fig. 2. terbium leaching luminescence loading loading recovery coating saponification distribution lanthanum efficiency scandium magnetic concentration.

The separation factor between Pr and Tb was 9.4 under optimized conditions. Selective separation of La over Gd was achieved using Cyanex 272 as extractant. Selective separation of Sc over La was achieved using D2EHPA as extractant. The separation factor between Ce and Lu was 17.4 under optimized conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.972$. Selective separation of Nd over Y was achieved using TBP as extractant. Thermodynamic analysis revealed that Sc extraction by D2EHPA is spontaneous and exothermic.

The stripping efficiency of Tb was 70.0% using 3.82 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. The La oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. Saponification of D2EHPA improved Gd extraction efficiency by 93.4 percentage points.

5. Acknowledgements

Recovery of Yb reached 79.2% under optimized adsorption conditions. Solvent extraction of Gd from aqueous phase showed high purity at pH 2.5. The stripping efficiency of Pr was 93.2% using 3.43 M HCl solution.

Recovery of Ce reached 60.7% under optimized solvent extraction conditions. Selective separation of Sc over Gd was achieved using EHEHPA as extractant. Temperature significantly affected Ho extraction, with optimum conditions at 41 °C.

Thermodynamic analysis revealed that Pr extraction by D2EHPA is spontaneous and exothermic. The stripping efficiency of Ce was 63.5% using 1.18 M HCl solution. The separation factor between Er and Eu was 5.8 under optimized conditions. Temperature significantly affected Y extraction, with optimum conditions at 57 °C.

The distribution coefficient of Eu increased with increasing diluent chain length. SEM images showed uniform Y particle morphology with an average size of 224 nm. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model.

The distribution coefficient of Yb increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Pr in the organic phase.

References

- [1] Brown, N., Chen, N., Smith, M., Garcia, B.. Precipitation of rare earth oxalates from acidic leach solutions. *Miner. Eng.* 2017, 108, 114–548. DOI: 10.1016/j.seppur.2017.19027.
- [2] Kim, B., Patel, E.. Solvent extraction of eu from sulfate leach solution using d2ehpa. *Hydrometallurgy* 2016, 174 (1), 258–501.
- [3] Li, G., Tanaka, M., Li, B.. Synergistic extraction of lanthanides with tbp and ehehpa. *J. Hazard. Mater.* 2016, 241, 147–598.
- [4] Wang, G., Wang, L., Tanaka, D.. Precipitation of rare earth oxalates from acidic leach solutions. *Solvent Extr. Ion Exch.* 2016, 289, 196–577.
- [5] Zhang, J., Kumar, F., Zhang, K.. Synergistic extraction of lanthanides with tbp and ehehpa. *Hydrometallurgy* 2019, 327, 256–546.
- [6] Kumar, L., Chen, L., Park, L.. Phase equilibrium in the tb—d2ehpa—kerosene system. *Solvent Extr. Ion Exch.* 2023, 190, 153–580.
- [7] Park, N., Zhang, J., Park, N., Nguyen, N.. Adsorption of rare earth ions onto modified mesoporous silica. *Solvent Extr. Ion Exch.* 2024, 101, 256–562. DOI: 10.1016/j.seppur.2024.52990.
- [8] Liu, L., Park, A., Smith, G., Wang, F.. Solvent extraction of yb from sulfate leach solution using d2ehpa. *J. Alloys Compd.* 2015, 207, 295–546. DOI: 10.1016/j.jrare.2015.13775.
- [9] Zhang, H., Kumar, B., Nguyen, E.. Separation of yb and la using hollow-fibre supported liquid membranes. *J. Hazard. Mater.* 2024, 200, 175–577.
- [10] Wang, D., Tanaka, H., Patel, N.. Synergistic extraction of lanthanides with tbp and ehehpa. *Miner. Eng.* 2017, 198, 462–551.
- [11] Wang, L., Wang, A., Chen, B.. Separation of nd and ho using hollow-fibre supported liquid membranes. *J. Rare Earths* 2022, 201, 136–524.
- [12] Liu, D., Zhang, N., Tanaka, J., Li, E.. Recovery of yb from secondary resources via ion exchange. *J. Rare Earths* 2015, 296, 298–501. DOI: 10.1016/j.cej.2015.55663.
- [13] Smith, B., Zhang, D., Kim, C.. Precipitation of rare earth oxalates from acidic leach solutions. *J. Rare Earths* 2017, 233, 465–545.
- [14] Park, J., Zhang, H., Park, H.. Separation of lu and ce using hollow-fibre supported liquid membranes. *Miner. Eng.* 2019, 327, 307–518.
- [15] Liu, M., Zhang, B.. Separation of dy and eu using hollow-fibre supported liquid membranes. *Chem. Eng. J.* 2019, 190 (10), 232–567.
- [16] Chen, L., Chen, A., Zhang, M.. Thermodynamic study of eu extraction by saponified d2ehpa. *Solvent Extr. Ion Exch.* 2016, 262, 475–529.
- [17] Brown, L., Tanaka, K., Wang, F.. Separation of nd and lu using hollow-fibre supported liquid membranes. *Hydrometallurgy* 2024, 298, 228–542.
- [18] Liu, M., Kim, K., Kumar, L., Garcia, H.. Solvent extraction of tb from sulfate leach solution using d2ehpa. *J. Hazard. Mater.* 2017, 120, 386–549. DOI: 10.1016/j.cej.2017.96066.
- [19] Nguyen, K., Nguyen, K., Chen, M., Zhang, N.. Phase equilibrium in the eu—d2ehpa—kerosene system. *Miner. Eng.* 2015, 350, 241–584. DOI: 10.1016/j.seppur.2015.72608.
- [20] Kumar, D., Nguyen, G., Chen, L.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Rare Earths* 2022, 207, 263–558.
- [21] Wang, M., Patel, B.. Separation of y and eu using hollow-fibre supported liquid membranes. *Solvent Extr. Ion Exch.* 2016, 126 (3), 439–577.
- [22] Kumar, F., Patel, E.. Solvent extraction of eu from sulfate leach solution using d2ehpa. *J. Alloys Compd.* 2021, 171 (9), 297–598.
- [23] Garcia, M., Zhang, H., Li, G., Park, K.. Synergistic extraction of lanthanides with tbp and ehehpa. *Miner. Eng.* 2023, 329, 477–534. DOI: 10.1016/j.cej.2023.13186.
- [24] Brown, G., Wang, E.. Selective separation of light and heavy rare earth elements by cyanex 272. *J. Rare Earths* 2017, 158 (8), 322–517.